

FIT5047: Assignment 1 Installation Guide

Version 1.1

This document provides supporting documentation to help you come to grips with the Pac-Man game environment which we will use for Assignment 1.

0. Required Skills

- **Programming with Python**
- **Using git to manage codes.**
- **Using command line interface.**

No worries if some of these skills are new to you - that's totally okay. These links right here are your pathway to learning. They're here to guide you at your own pace, leading you step-by-step towards mastery. Dive in and out of these resources as you feel appropriate.

- **Python introduction:**
https://www.w3schools.com/python/python_intro.asp
- **Git introduction:**
<https://github.com/git-guides>
- **Linux/Mac CLI (Command Line Interface) Guide:**
<https://ubuntu.com/tutorials/command-line-for-beginners#1-overview>
- **Windows CMD cheat sheet:**
<http://www.cs.columbia.edu/~sedwards/classes/2015/1102-fall/Command%20Prompt%20Cheatsheet.pdf>
- **The MIT Missing Semester in Computer Science:**
<https://missing.csail.mit.edu/>

1. Installation and Setup

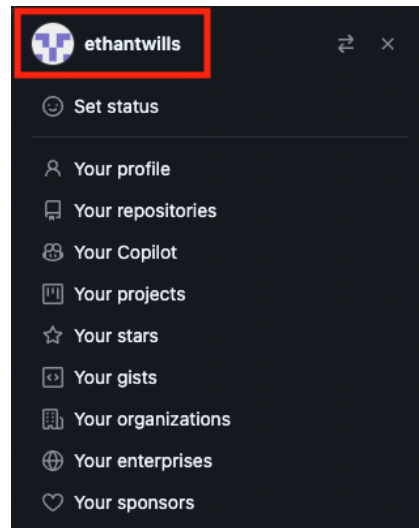
(a) Create a GitHub account

For this assignment you will upload your code to a GitHub repository from where it will be automatically evaluated.

First, create a [GitHub account](#) using your **Monash email address**. Once you've created your account, find your username by clicking on your profile image icon in the top right of the screen.



Your username should be at the top of the sidebar that appears, for example:



(b) Register for the auto-grader server:

Our auto-grading server evaluates your submissions and lets you compare your progress in the assignment against your classmates, using a live leaderboard. You can find the auto-grader website here: <http://fit5047s1.rep.monash.edu/>.

NOTE: if you have trouble accessing the server from home, you may need to connect to the Monash VPN. Instructions for that are available [here](#).

From the auto-grader website, you will need to register another account using your **Monash email address** and your **GitHub username**. You will then receive an email invitation to join a private Git repository.

The invitation email will be sent to the email address associated with your GitHub username (this should be your Monash email address if you followed the instructions above). Accept the email invitation to get access to your repository.

WARNING: If you don't use the correct GitHub username you won't get an invitation.

This repository will contain all the code you need for your assignment.

PLEASE NOTE:

If you complete this setup before the assignment 1 code is released, your repository will be empty. Don't worry, we'll push the code to the **assignment1** branch of your repo when it is ready. During the assignment period you will push your changes to this branch for evaluation.

You can now safely proceed to the next step.

(b) Create a development environment

For this assignment we will implement our agents using the **Python** programming language. We also recommended using **miniconda**, a lightweight tool that helps us install Python, and manage different Python environments.

Follow [this guide](#) to install miniconda. Please refer to the appropriate instructions for your operating system. Alternatively, you can use [anaconda](#) which provides more features than miniconda, such as a GUI interface.

Once miniconda is installed, type the following commands into your terminal:

```
> conda create -n fit5047_pacman python=3.9
> conda activate fit5047_pacman
> conda install -c anaconda tk
```

(c) Install Git

If you don't already have the git version management software installed, you will need to do that now. The easiest approach is to install git via miniconda:

```
> conda install anaconda::git
```

Confirm that git successfully installed by checking its version

```
> git --version
```

At the time of writing the latest version was 2.45.2. Your version number may be different.

(d) Clone the repo to your computer:

Use the terminal and navigate to a directory in which you would like to clone the Pac-Man repository. For example, from your home directory:

```
> mkdir fit5047
> cd fit5047
```

For a refresher on navigating with a terminal interface, you may like to consult the [MIT Missing Semester in Computer Science](#), which is a terrific resource for learning not just about the terminal interface, but also the `git` and a variety of command line tools.

Next, from the “source” page of your GitHub repository, click the `<> Code ▾` button, copy the “Clone” command shown and paste it into the terminal. The command should look something like this:

```
> git clone https://github.com/FIT5047-2025/<YOUR_USERNAME>.git
```

If you use HTTPS to clone you will be prompted to enter your GitHub username and password. Alternatively, you can use SSH by [creating a key](#) and [adding it to your GitHub account](#), or by [creating a GitHub access token](#).

Once the command finishes you will have a new subdirectory called `<YOUR_USERNAME>`, in your current directory. For now the directory will only contain a `README.md` file.

(e) Access the `assignment1` branch

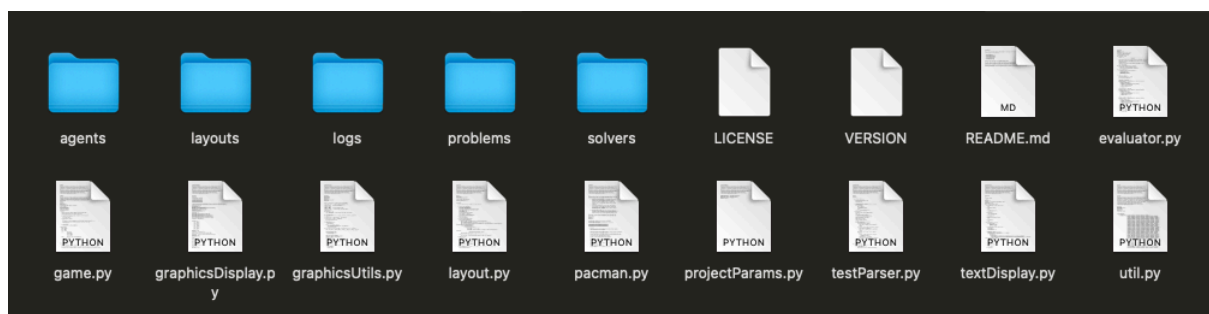
Navigate to the directory where you cloned your repo and run the following command.

```
> git pull
```

This will pull/download the `assignment1` branch to your local repo. Next, run the following command.

```
> git checkout assignment1
```

This will switch you to the `assignment 1` branch, and the contents of the `<YOUR_USERNAME>` directory should now look like this:



The `agents`, `layouts`, `problems` and `solvers` directories are all important for you. We will go through each of these directories in more detail in Section 3. **You shouldn't modify any of the python files in this directory.**

(f) Additional support software (OPTIONAL)

Many programmers like to write code in an Integrated Development Environment (IDE). If you want to try an IDE but aren't sure which, we suggest [VS Code](#).

Another popular (optional) tool is [Jupyter notebooks](#), which you can use for your own testing and development. Note that all files you submit must be the specified .py files (i.e., please do not submit your Jupyter notebook!).

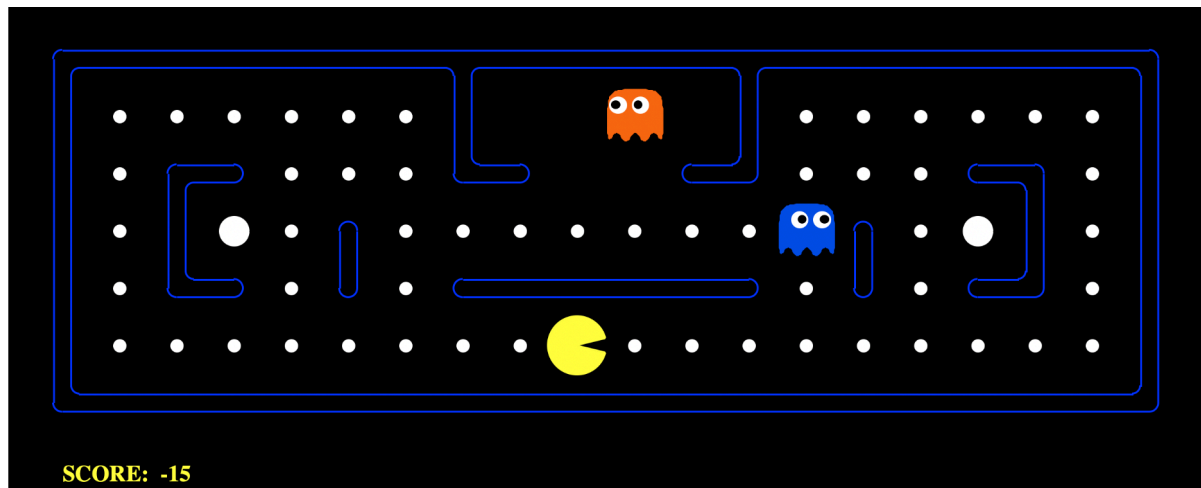
2. Test your setup

Once you have completed the above Instructions we make sure the environment is running correctly. We want to bring up the visualiser and run the game with two commands to verify the installation was successful. Make sure you are in the directory where the `pacman.py` file is located then enter the commands below in the terminal. Each game will end when Pac-Man has eaten all the food on the map, or when Pac-Man is eaten by a ghost.

Demo with a keyboard controller:

```
> python pacman.py -l smallClassic.lay
```

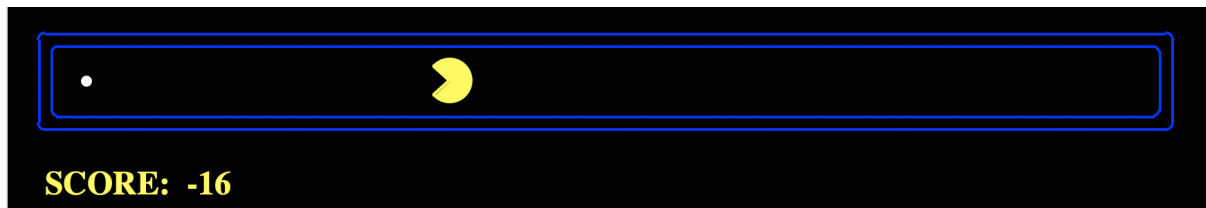
This command will begin a new game and allow you to control Pac-Man with the WASD keys on your keyboard.



Demo with a very simple agent:

```
> python pacman.py --layout testMaze.lay --pacman GoWestAgent
```

This command will begin a new game where the automated controller only ever tells Pac-Man to “Go West”.



You can use the `-o` flag to specify the file name for saving a log file in the **logs** directory. You can inspect these log files for debugging purposes but you should not modify them. More information about the contents of the log files is contained in Section 6.

3. Next Steps

- Read the Assignment Specification and Marking Rubric (on Moodle)
- Read the Marking Rubric again, and make sure you really understand what is expected.
- Check out the “Getting Started Guide” (on Moodle) for descriptions of the Pac-Man game environment and tips for how to get started with the assignment.

Revision History

1.1

- Updated server URL
- Added a note about Monash VPN